Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/01/2007
PI Name:	Nalette, Timothy Ph.D.		
Project Title:	Advanced CO2, Humidity, and Trace Contaminant Control System for Closed Loop Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	t.nalette@hs.utc.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	860-654-2403
Organization Name:	Hamilton Sundstrand Space Systems International, Inc.		
PI Address 1:	1 Hamilton Road 1A-2-W59		
PI Address 2:			
PI Web Page:			
City:	Windsor Locks	**State:**	CT
Zip Code:	06096-1010	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	08/11/2004	**End Date:**	10/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2008 per PI (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Papale, William (Hamilton Sundstrand)		
Grant/Contract No.:	NNJ04HF73A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Conduct basic research associated with amine based sorbents for combined humidity and carbon dioxide control for closed loop life support. The basic research will investigate alternate sorbent formulations and configurations that will address enhanced kinetics and loadings to enhance overall system performance. These enhancements will result in reduced system mass, volume and power, while also enhancing overall vehicle integration issues associated with the addition of the humidity control function. Based on the results of the basic sorbent research, we will design, fabricate, assemble and test a full scale, deliverable prototype system.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	May have applicability to CO2 capture from stationary power plants to address greenhouse gas abatement.
Task Progress:	The primary achievements for the time period of August 2006-September 2007 include – completed manufacturing, assembly and testing of a second, full scale CEV sized system configured for sweep gas regeneration; delivery of the second, full scale prototype to NASA JSC; a valve design study investigating an alternate rotary cylinder concept; CO2 and H2O isotherm testing of SA9T sorbent to support SIMA analysis; and continued research activities on solid amine properties and performance at the University of Hartford. Field support by Hamilton Sundstrand personnel for both prototype units was also provided for the initial start-up and operation of the systems at NASA-JSC. The report documentation includes performance data for the second prototype unit, SA9T isotherm testing report and presentation materials, subcontractor reports and 2007 ICES papers and presentation materials.
Bibliography Type:	Description: (Last Updated: 02/19/2009)
Papers from Meeting Proceedings	Nalette T, Papale W, Smith F. "Development Status of the Carbon Dioxide and Moisture Removal Amine Swing-bed (CAMRAS)." Presented at the 37th International Conference on Environmental Systems in Chicago, IL, July 2007. ICES paper 2007-01-3157, 5 p., July 2007. , Jul-2007